Knowledge Base: LLD

Purpose

This document describes the low-level design of the Knowledge Base feature in the DAY2 platform. It covers the system architecture, component interactions, data models, and API contracts required to implement a RAG-based knowledge management system enabling MSPs to upload, organize, and query tenant-specific documents using natural language.

This document serves as the technical reference for the engineering team during implementation.

Flow diagram

Upload document flow

Upload document flow

Could not identify user

Document Ingestion flow

KB Ingestion Flow

Could not identify user

Delete Document

Delete Doc Flow

Could not identify user

Query Knowledge Base

Query Knowledge Base Flow

Could not identify user

Delete Collection

Delete Collection

Could not identify user

API Schema

Base URL: /api/v1/

HTTP Status Codes

Code	Description
200	Success
201	Created
204	No Content (Delete)
400	Validation Error
403	Access Denied
404	Not Found
409	Conflict (Duplicate)
422	Unprocessable Entity
500	Internal Server Error

Enums

- Category: tenant, workload

Response structure

Success response structure

```
{
  "success": true,
  "message": "Collections retrieved successfully",
  "data": [],
  "meta": {
    "total": 45,
    "page": 1,
    "page_size": 20,
    "total_pages": 3
  }
}
```

Error response structure

```
{
  "success": false,
  "message": "Validation failed",
```

```
  "errors": [
    {
      "field": "name",
      "message": "Collection name is required"
    },
    {
      "field": "category",
      "message": "Invalid category value. Allowed values: workload, tenant"
    }
  ],
  "data": null
}
```


Collections Endpoints

Method	Endpoint	Description
POST	/collections	Create a collection
GET	/collections	List Collections
GET	/collections/{id}	Get collection along with documents
PATCH	/collections/{id}	Update a collection
DELETE	/collections/{id}	Delete a collection and related documents
GET	/collections/summary	Returns the total number of collections and number of workload/Teanant collection

Create collection

Request Payload


```
{
  "customer_id": "550e8400-e29b-41d4-a716-446655440000",
  "tenant_id": "7c9e6679-7425-40de-944b-e07fc1f90ae7",
  "name": "Production Workloads",
  "description": "Collection of all production workload resources",
  "category": "tenant",
  "created_by": "3fa85f64-5717-4562-b3fc-2c963f66afa6"
}
```


Response


```
{
```

```
  "success": true,
  "message": "Collection created successfully",
  "data": {
    "id": "550e8400-e29b-41d4-a716-446655440001",
    "customer_id": "550e8400-e29b-41d4-a716-446655440000",
    "tenant_id": "7c9e6679-7425-40de-944b-e07fc1f90ae7",
    "name": "Production Workloads",
    "description": "Collection of all production workload resources",
    "category": "tenant",
    "documents_count": 0,
    "created_at": "2026-06-02T10:00:00Z",
    "created_by": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
    "updated_at": "2026-06-02T10:00:00Z",
    "updated_by": "3fa85f64-5717-4562-b3fc-2c963f66afa6"
  },
  "meta": null
}
```

Get collections

Request with filters

```
GET /collections?category=tenant
  &search=Production
  &sort_by=created_at
  &order=asc
  &page=1
  &page_size=20
```

Response with filters

```
{
  "success": true,
  "message": "Collection created successfully",
  "data": [{
    "id": "550e8400-e29b-41d4-a716-446655440001",
    "customer_id": "550e8400-e29b-41d4-a716-446655440000",
    "tenant_id": "7c9e6679-7425-40de-944b-e07fc1f90ae7",
    "name": "Production Workloads",
    "description": "Collection of all production workload resources",
    "category": "tenant",
    "documents_count": 4,
    "created_at": "2026-06-02T10:00:00Z",
    "created_by": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
    "updated_at": "2026-06-02T10:00:00Z",
    "updated_by": "3fa85f64-5717-4562-b3fc-2c963f66afa6"
  },{
    "id": "550e8400-e29b-41d4-a716-446655440001",
    "customer_id": "550e8400-e29b-41d4-a716-446655440000",
    "tenant_id": "7c9e6679-7425-40de-944b-e07fc1f90ae7",
    "name": "Production Workloads",
    "description": "Collection of all production workload resources",
    "category": "tenant",
    "documents_count": 2,
    "created_at": "2026-06-02T10:00:00Z",
    "created_by": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
    "updated_at": "2026-06-02T10:00:00Z",
    "updated_by": "3fa85f64-5717-4562-b3fc-2c963f66afa6"
  }
  ],
```

```
  "meta": {
    "total": 45,
    "page": 1,
    "page_size": 20,
    "total_pages": 3
  }
}
```

Update collection

Request

```
{
  "name": "Production Workloads - Updated",
  "description": "Updated collection of all production workload
resources",
  "updated_by": "3fa85f64-5717-4562-b3fc-2c963f66afa6"
}
```

Response

```
{
  "success": true,
  "message": "Collection created successfully",
  "data": {
    "id": "550e8400-e29b-41d4-a716-446655440001",
    "customer_id": "550e8400-e29b-41d4-a716-446655440000",
    "tenant_id": "7c9e6679-7425-40de-944b-e07fc1f90ae7",
    "name": "Production Workloads - Updated",
    "description": "Updated collection of all production workload
resources",
    "category": "tenant",
    "created_at": "2026-06-02T10:00:00Z",
    "created_by": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
    "updated_at": "2026-06-02T10:00:00Z",
    "updated_by": "3fa85f64-5717-4562-b3fc-2c963f66afa6"
  },
  "meta": null
}

```

Summary collections

Response

{ "total": 5, "byCategory": { "Tenant": 3, "Workload": 2 } }

Labels endpoints

Method	Endpoint	Description
POST	/labels	Create a label
GET	/labels	List labels

Create Label

Request


```json
{
  "name": "Critical Infrastructure",
  "customer_id": "550e8400-e29b-41d4-a716-446655440000",
  "tenant_id": "7c9e6679-7425-40de-944b-e07fc1f90ae7"
}
```


Response


```json
{
  "success": true,
  "message": "Label created successfully",
  "data": {
    "id": "770a9511-b41d-63f6-c938-668877662223",
    "name": "Critical Infrastructure",
    "customer_id": "550e8400-e29b-41d4-a716-446655440000",
    "tenant_id": "7c9e6679-7425-40de-944b-e07fc1f90ae7"
  },
  "meta": null
}
```


List labels

Response


```json
{
  "success": true,
  "message": "Labels retrieved successfully",
  "data": [
    {
      "id": "660b8400-e29b-41d4-a716-446655440011",
      "name": "Architecture",
      "customer_id": null,
      "tenant_id": null
    },
    {
      "id": "660b8400-e29b-41d4-a716-446655440012",
      "name": "Compliance",
      "customer_id": null,
      "tenant_id": null
    },
    {
      "id": "770a9511-b41d-63f6-c938-668877662223",
      "name": "Critical Infrastructure",
      "customer_id": "550e8400-e29b-41d4-a716-446655440000",
      "tenant_id": "7c9e6679-7425-40de-944b-e07fc1f90ae7"
    }
  ]
}
```


Documents Endpoints

Method	Endpoint	Description

POST	/collections/{collection_id}/documents/upload_url	Generates s3 presigned urls for uploading documents
GET	/collections/{collection_id}/documents	List documents in a collection
GET	/collections/{collection_id}/documents/{document_id}	Get document along with a presigned url to view the document
PATCH	/collections/{collection_id}/documents/{document_id}	Update document information
DELETE	/collections/{collection_id}/documents/{document_id}	Trigger async process to delete document from AWS Knowledge Base, S3 and database
POST	/collections/{collection_id}/documents/{document_id}/summarization	Retry summary generation

Genearte upload urls

Request

```
{
  "customer_id": "550e8400-e29b-41d4-a716-446655440000",
  "tenant_id": "7c9e6679-7425-40de-944b-e07fc1f90ae7",
    "name": "architecture.pdf",
    "size": 2048576,
    "type": "pdf",
    "additional_context": "Q2 2026 Architecture Review",
    "label_ids": [
      "660b8400-e29b-41d4-a716-446655440011",
      "660b8400-e29b-41d4-a716-446655440012"
    ],
    "created_by": "3fa85f64-5717-4562-b3fc-2c963f66afa6"
}
```

Response

```
{
  "success": true,
  "message": "Upload URLs generated successfully",
```

```
  "data": {
      "document_id": "b5a1d5e1-210f-487f-b568-24b750ea7b55",
      "name": "architecture.pdf",
      "s3_key": "550e8400/7c9e6679/660f9511/b5a1d5e1/architecture.pdf",
      "presigned_url": "https://s3.amazonaws.com/bucket/550e8400/.../architecture.pdf?X-Amz-Signature=...",
      "presigned_url_expiry": "2026-06-02T10:15:00Z",
      "status": "initiated"
  },
  "meta": null
}
```

Get documents

Request with filters

```
GET /collections/660f9511-f30c-52e5-b827-557766551112/documents
  ?status=available
  &label_ids=c6b2e6f2-321f-598a-c679-35c861fb8c66,c6b2e6f2-321f-598a-c679-35c861fb8c66
  &name=
  &description=
  &user_id=c6b2e6f2-321f-598a-c679-35c861fb8c66,c6b2e6f2-321f-598a-c679-35c861fb8c66
  &order=desc
  &page=1
  &page_size=20
```

Response

```
{
  "success": true,
  "message": "Document retrieved successfully",
  "data": [{
    "id": "b5a1d5e1-210f-487f-b568-24b750ea7b55",
    "customer_id": "550e8400-e29b-41d4-a716-446655440000",
    "tenant_id": "7c9e6679-7425-40de-944b-e07fc1f90ae7",
    "collection_id": "660f9511-f30c-52e5-b827-557766551112",
    "name": "architecture.pdf",
    "size": 2048576,
    "type": "pdf",
    "additional_context": "Q2 2026 Architecture Review",
    "summary": "This document outlines the Q2 architecture review...",
    "s3_key": "550e8400/7c9e6679/660f9511/b5a1d5e1/architecture.pdf",
    "label_ids": [
      "660b8400-e29b-41d4-a716-446655440011",
      "660b8400-e29b-41d4-a716-446655440012"
    ],
    "status": "indexed",
    "created_at": "2026-06-02T10:00:00Z",
    "created_by": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
    "updated_at": "2026-06-02T10:05:00Z",
    "updated_by": "3fa85f64-5717-4562-b3fc-2c963f66afa6"
  },{
    "id": "b5a1d5e1-210f-487f-b568-24b750ea7b55",
    "customer_id": "550e8400-e29b-41d4-a716-446655440000",
    "tenant_id": "7c9e6679-7425-40de-944b-e07fc1f90ae7",
    "collection_id": "660f9511-f30c-52e5-b827-557766551112",
    "name": "architecture.pdf",
```

```json
      "size": 2048576,
      "type": "pdf",
      "additional_context": "Q2 2026 Architecture Review",
      "summary": "This document outlines the Q2 architecture review...",
      "s3_key": "550e8400/7c9e6679/660f9511/b5a1d5e1/architecture.pdf",
      "label_ids": [
        "660b8400-e29b-41d4-a716-446655440011",
        "660b8400-e29b-41d4-a716-446655440012"
      ],
      "status": "indexed",
      "created_at": "2026-06-02T10:00:00Z",
      "created_by": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
      "updated_at": "2026-06-02T10:05:00Z",
      "updated_by": "3fa85f64-5717-4562-b3fc-2c963f66afa6"
    }
  ],
  "meta": {
    "total": 45,
    "page": 1,
    "page_size": 20,
    "total_pages": 3
  }
}
```

Get document

Request

```
GET /collections/660f9511-f30c-52e5-b827-557766551112/documents/b5a1d5e1-210f-487f-b568-24b750ea7b55
```

Response

```json
{
  "success": true,
  "message": "Document retrieved successfully",
  "data": {
    "id": "b5a1d5e1-210f-487f-b568-24b750ea7b55",
    "customer_id": "550e8400-e29b-41d4-a716-446655440000",
    "tenant_id": "7c9e6679-7425-40de-944b-e07fc1f90ae7",
    "collection_id": "660f9511-f30c-52e5-b827-557766551112",
    "name": "architecture.pdf",
    "size": 2048576,
    "type": "pdf",
    "additional_context": "Q2 2026 Architecture Review",
    "summary": "This document outlines the Q2 architecture review...",
    "s3_key": "550e8400/7c9e6679/660f9511/b5a1d5e1/architecture.pdf",
    "presigned_url": "https://s3.amazonaws.com/bucket/550e8400/.../architecture.pdf?X-Amz-Signature=...",
    "presigned_url_expiry": "2026-06-02T10:15:00Z",
    "label_ids": [
      "660b8400-e29b-41d4-a716-446655440011",
      "660b8400-e29b-41d4-a716-446655440012"
    ],
    "status": "indexed",
    "created_at": "2026-06-02T10:00:00Z",
    "created_by": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
    "updated_at": "2026-06-02T10:05:00Z",
```

```json
    "updated_by": "3fa85f64-5717-4562-b3fc-2c963f66afa6"
  },
  "meta": null
}
```

Update document

Request

```json
{
  "name": "architecture-v2.pdf",
  "additional_context": "Updated Q2 2026 Architecture Review",
  "label_ids": [
    "660b8400-e29b-41d4-a716-446655440011"
  ],
  "updated_by": "3fa85f64-5717-4562-b3fc-2c963f66afa6"
}
```

Response

```json
{
  "success": true,
  "message": "Document updated successfully",
  "data": {
    "id": "b5a1d5e1-210f-487f-b568-24b750ea7b55",
    "customer_id": "550e8400-e29b-41d4-a716-446655440000",
    "tenant_id": "7c9e6679-7425-40de-944b-e07fc1f90ae7",
    "collection_id": "660f9511-f30c-52e5-b827-557766551112",
    "name": "architecture-v2.pdf",
    "size": 2048576,
    "type": "pdf",
    "additional_context": "Updated Q2 2026 Architecture Review",
    "summary": "This document outlines the Q2 architecture review...",
    "s3_key": "550e8400/7c9e6679/660f9511/b5a1d5e1/architecture.pdf",
    "label_ids": [
      "660b8400-e29b-41d4-a716-446655440011"
    ],
    "status": "indexed",
    "created_at": "2026-06-02T10:00:00Z",
    "created_by": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
    "updated_at": "2026-06-02T11:00:00Z",
    "updated_by": "3fa85f64-5717-4562-b3fc-2c963f66afa6"
  },
  "meta": null
}
```

Delete document

Request

```
DELETE /collections/660f9511-f30c-52e5-b827-557766551112/documents/b5a1d5e1-210f-487f-b568-24b750ea7b55
```

Response

```json
{
  "success": true,
```

```json
  "message": "Document queued for deletion successfully",
  "data": {
    "id": "b5a1d5e1-210f-487f-b568-24b750ea7b55",
    "status": "deleting"
  },
  "meta": null
}

```

Retry summarization

Response

```json
{
  "success": true,
  "message": "Document updated successfully",
  "data": {
    "id": "b5a1d5e1-210f-487f-b568-24b750ea7b55",
    "status": "MetadataUploaded",
    "updated_at": "2026-06-02T11:00:00Z",
    "updated_by": "3fa85f64-5717-4562-b3fc-2c963f66afa6"
  },
  "meta": null
}
```

Query knowledge base

Method	Endpoint	Description
POST	/collections/{collection_id}/query	Send a message and streams response via SSE

Query threads

Headers

```
Content-Type: application/json
Accept: text/event-stream
```

Request

```json
{ "message": "What is the current architecture of the production workload?",
  "session_id": "2e10926c-1c34-4e62-b9ef-cd0d5a0ca86d"" }
```

Response

```
event: session
data: {"session_id": "2e10926c-1c34-4e62-b9ef-cd0d5a0ca86d"}

```

```
event: text
data: {"text": "The production"}

event: text
data: {"text": " workload follows"}

event: text
data: {"text": " a three-tier architecture..."}

event: citation
data: {
  "references": [
    {
      "document_id": "b5a1d5e1-210f-487f-b568-24b750ea7b55",
      "document_name": "production-architecture.pdf",
      "content_type": "IMAGE",
      "page_number": 3,
      "summary": "Architecture diagram showing the three-tier setup with CloudFront, ECS Fargate, ALB, Cognito and Aurora PostgreSQL across multiple availability zones."
    },
    {
      "document_id": "c6b2e6f2-321f-598a-c679-35c861fb8c66",
      "document_name": "infrastructure-runbook.pdf",
      "content_type": "TEXT",
      "page_number": 7,
      "summary": "Runbook section describing the ECS Fargate cluster configuration, ALB listener rules, and Cognito user pool setup for the production environment."
    }
  ]
}
}

event: done
data: {
  "message_id": "m2b3c4d5-e6f7-8a9b-0c1d-e2f3a4b5c6d7",
  "thread_id": "d4e5f6a7-b8c9-4d0e-a1b2-c3d4e5f6a7b8"
}
```

Database

What should it support?

- User should be able to search for documents by the following
  - Collection ID
  - Tenant ID
  - Status
  - MSPID
  - UserID
  - Name
  - Labels
  - Created By
  - Sort by Created At

- The system should support getting all documents that are in “Analysing”/”Deleting” status to fetch if the document has been indexed successfully by the Knowledge Base

Options

DynamoDB

Advantages

- Native Streams for event-driven flows
- Serverless-native, no connection management

Disadvantages

- We need to have multiple index to support the use case
  - For documents
    - One index with PK as Customer_ID#Tenant_ID and SK as Collection_ID and any other filters we have to do at the service level.
    - Another index with PK as status to support getting all documents that are in “Analysing”/”Deleting” status this creates a hot partition
- We need to filtering of Labels, Name, User, etc at application level.

PostgreSQL Serverless v2

Advantages

- Flexible querying
- We can apply filtering logic at database level.

Disadvantages

- Dosen’t natively support publishing events when data changes (this is useful to trigger follow up tasks such as triggering ingestion into knowledge base once the summary is generated)

ER Diagram

KB: ER Diagram

Could not identify user

We ran into a problem

We've noticed an issue that may affect your experience on Confluence. Select “Refresh” below to fix it.

Refresh

Schema

collections

A logical groups for documents scoped to a specific MSP and tenant.

Column Name	Type	PK / FK	Description
id	UUID4	PK	Unique identifier for the collection
customer_id	UUID4	-	Identifier of the owning MSP
tenant_id	UUID4	-	Identifier of the tenant this collection belongs to
name	VARCHAR(255)	-	Display name of the collection; Unique per tenant
status	VARCHAR(50)	-	Needed to mark that a collection is marked for deletion
description	TEXT	-	Optional description of the collection
category	INT	-	enum field for workload/tenant
documents_count	INT	-	Number of documents that belong to the Collection

created_at	TIMESTAMPTZ	-	Timestamp when the collection was created
created_by	UUID4	-	User ID of the MSP user who created the collection
updated_at	TIMESTAMPTZ	-	Timestamp of the last update
updated_by	UUID4	-	User ID of the MSP who last updated the collection

Composite Index: customer_id, tenant_id

documents

Stores metadata for documents uploaded by MSP users, including S3 location, and processing status.

Column Name	Type	PK / FK	Description
id	UUID4	PK	Unique identifier for the document
customer_id	UUID4	-	Identifier of the owning MSP;
tenant_id	UUID4	-	Identifier of the tenant this document belongs to
collection_id	UUID4	FK → collections.id	Collection this document belongs to

name	VARCHAR(255)	-	Display name of the document; unique per collection
size	BIGINT	-	File size in bytes
type	VARCHAR(50)	-	File type e.g. pdf, docx, txt
additional_context	TEXT	-	Optional free-text context provided by the MSP user, Can be NULL
summary	TEXT	-	Auto-generated summary of the document
s3_key	VARCHAR(100)	-	S3 object key used to locate the document in the storage bucket; follow the pattern `{customer_id}/{tenant_id}/{collection_id}/{document_id}/{filename}`
status	VARCHAR(50)	-	Document lifecycle: Pending, MetadataUploaded, Summarised, Indexed, Deleting, Failed

created_at	TIMESTAMPTZ	-	Timestamp when the document record was created
created_by	UUID4	-	User ID of the MSP user who uploaded the document
updated_at	TIMESTAMPTZ	-	Timestamp of the last update
updated_by	UUID4	-	User ID of the MSP user who last updated the document

Composite Index: customer_id, tenant_id, collection_id

document_state_transitions

Stores the state transitions for the document

Column Name	Type	PK / FK	Description
id	UUID4	PK	Unique identifier for the system label
document_id	UUID4	FK	document it belongs to
from_status	VARCHAR(50)	-	Original state the document was in.
to_status	VARCHAR(50)	-	New state the document was in.

created_at	TIMESTAMPTZ	-	Timestamp of the state transition

Index: document_id

labels

Stores platform-defined labels available to all MSPs along with tenant specific labels

Column Name	Type	PK / FK	Description
id	UUID4	PK	Unique identifier for the system label
name	VARCHAR(100)	-	Unique name of the system label e.g. Architecture, Compliance
customer_id	UUID4	-	Identifier of the MSP that created this label; NULL means system label
tenant_id	UUID4	-	Identifier of the Tenant that this label belongs to; NULL means system label

Composite Index: customer_id, tenant_id

document_labels

Join table that associates documents with labels; supports many-to-many label assignment.

Column Name	Type	PK / FK	Description
document_id	UUID4	PK, FK → documents.id	Reference to the labelled document
label_id	UUID4	PK, FK → labels.id	Reference to either a labels

S3 Buckets

- knowledge_base_docs to store all knowledge base
- knowledge_base_multimedia to store multimedia assets like images, etc during the embedding process
- knowledge_base_vectors to store vector embeddings.
- knowledge_base_sessions to be used for Session management for strands

Day2 Events

domain: marvinknowledgebase

Entity	**Action**
Collection	Created
	CreateFailed
	Updated
	UpdateFailed
	MarkedForDeletion
	MarkedForDeletionFailed
	Deleted
	DeleteFailed
Document	Created

	CreateFailed
	MetadataUploaded
	MetadataUploadFailed
	Updated
	UpdateFailed
	Summarized
	SummarizationFailed
	Ingested
	IngestionFailed
	MarkedForDeletion
	MarkedForDeletionFailed
	Deleted
	DeletionFailed
Label	Created
	CreationFailed

Use Cases

Use Case	Describe the flow
When User logins he should see a welcome screen	This will be handled in frontend using local browser storage
User should be able to create collections and categorise them as workload/tenant	Call POST for collections endpoint
User should be able to filter collections based on Name,	Call GET for /collections

Description and Category. Should be able to sort by Created At	
User should be able to update collection details	Call PATCH for collections/{collection_id}
User should be able to upload documents to a collection	Upload document flow
User should be able to update document details	Call PATCH for collections/{collection_id}/documents/{document_id}
Once the document is uploaded user should be able to see the summary of the document	KB Ingestion Flow
The document should be ingested into the knowledge base	KB Ingestion Flow
User should be able to retry summary generation if it fails	Invoke Retry endpoint on documents
User should be able to query knowledge base	Query Knowledge Base Flow
User should be able to delete a document from collection	Delete Doc Flow
User should be able to Query a collection	Query Knowledge Base Flow
User should be able to Delete a collection	Delete Collection
Deleting a Tenant	Out of Scope for v1

Evaluation

AWS Bedrock Evaluation assesses our Knowledge Base pipeline responses using the following metrics

Quality metrics (7 available)

Evaluate the effectiveness of generated responses.

Metric	Description	Useful or not
Helpfulness	Measures how useful and comprehensive responses are in answering questions.	Useful
Correctness	Measures the accuracy of responses in answering questions.	Useful
Logical Coherence	Measures whether responses lack logical gaps, inconsistencies, or contradictions.	Not useful for RAG pipeline; correctness suffices.
Faithfulness	Measures factual accuracy and avoidance of hallucinations.	Useful
Completeness	Measures how fully responses answer all parts of questions.	Useful
Citation Precision	Measures accuracy of cited passages.	Useful
Citation Coverage	Measures extent responses are supported by cited passages and flags missing citations.	Useful

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Responsible AI metrics (3 available)

Evaluate appropriateness and safety of generated responses.

Metric	Description

Harmfulness	Measures harmful content involving hate, insult, or violence.
Refusal	Measures evasiveness in refusing to answer questions.
Stereotyping	Measures generalized statements about individuals or groups.

We can run these checks at build time to detect any drop in metric quality.

Testability